UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2025/2026

UECS3253 WIRELESS APPLICATION DEVELOPMENT

Personal Finance Tracker

Group Name:			
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PART		**CO**	**Component**	**Criteria**	**Marks Allocated**
A	1	CO2	Report	Screens & Layout	/5
	2			Use of Appropriate Components	/5
	3			Reusability (Custom Components) inserts/updates	/5
	4			Implementation of React-Navigation: Stack, Drawer, Tab	/10
	5		Demonstration	User Interface	/5
Subtotal					**/30**
B	1	CO3	Report	Appropriate Usage of Data Persistence	/10
	2			Implementation of Data Persistence	/10
	3			CRUD Operations & input validation	/10
	4		Demonstration	CRUD and Data Persistence	/5
Subtotal					**/35**
C	1	CO4	Report	Appropriate Usage of Cloud Connectivity	/15
	2			Implementation of Cloud Connectivity	/15
	3		Demonstration	Cloud Connectivity	/5
Subtotal					**/35**

[Total: /100 marks]

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BRIEF SUMMARY

The Personal Financial Tracker is a comprehensive, cross-platform mobile application developed using React Native. Designed with a modern, minimalist and aesthetic style, this application aims to empower users to seamlessly track, manage, and analyse their daily income and expenses. By integrating intuitive data entry forms, real-time cloud synchronization, and office accessibility, this app also provides a highly responsive and secure financial management experience. The primary objective of this app is to offer users a clear, visual representation of their financial health, enabling better budgeting and financial decision-making on the go.

KEY FEATURES

This application incorporates several advanced technical and user-centric features:

Core Financial Management:

- **Comprehensive CRUD Operations:**
  Users can effortlessly create, read, update, and delete their transaction records directly with instantaneous UI feedback.
- **Multi-Wallet Management:**
  This app provides multiple wallets to view user financial health globally or drill down into specific wallets. User also can create and manage multiple wallets with custom names and icons to keep their different financial streams separate.
- **Smart Budgeting**
  This app will provide a visual progress bars that show how much of your monthly budget you've consumed. When it exceeds 85% of user allocated spending, the progress colour will change to red to alert the user the “Near Budget” overrun.
- **Comprehensive Transaction Management**
  Each transaction will be record and display at the full searchable history of all transactions with filtering capabilities. Each transaction can also be categorized using a pre-defined set of vibrant, easy-to-read categories and emojis.

Advanced Insights & Analytics:

- **Financial Calendar:**
  A dedicated analytics dashboard featuring a calendar that show spending of each day and a dynamic pie chart that automatically aggregate and categorize spending data for deeper financial insights.
- **Real-Time Cloud Synchronization:**
  Powered by a robust backend, ensuring that users' financial data and wallets are securely backed up and synchronized continuously to prevent data loss.
- **Dynamic UI/UX:**
  The app would also provide a fully integrated theme system that adapts the UI for comfortable viewing in both light and dark environments. It's powered by ***react-native-reanimated*** and ***Animated*** API for smooth transition and interactive feedback.

Security & Technical Hightlights:

- **Enterprise-Grade Security:**
  A dedicated "Secure Access" layer that protects your financial data from unauthorized local access by using the security PIN lock. Besides that, secure session management via JWT tokens to ensure your profile and data remain private.
- **Data Presistence & Offline Fallback:**
  This app will be utilizing ***AsyncStorage,*** **the application caches recent transaction history,** allowing users to view their financial records even without an active internet connection
- **Optimized Performance:**
  The frontend utilizes React Native's ***InteractionManager*** to ensure lag-free transitions and animations.

Part A

SCREENS, COMPONENTS AND REACT-NAVIGATION

A.1 Screens & Layout

The Personal Financial Tracker application is designed with a premium, dark-themed Neo-Banking aesthetic, prioritizing readability, focus and a seamless user experience. The structural layout of each screens is engineered to be highly responsive, adapting smoothly to various mobile device dimensions and screen ratios.

1. **Flexbox-Driven Responsive Layout**

   The layout structure is relies on React Nativc's ***Flexbox*** model. By utilizing properties such as ***flex: 1*** , ***justifyContent***, and ***alignItems,*** the application ensures that UI elements such as the total balance summary cardds, transaction list rows, and the input forms are dynamically proportioned. This prevents the UI conflicts and ensures the consistent visual hierarchy across both small and large devices.

2. **Component Grouping and Card Interface**

   To manage complex financial data visually, the layout emplys a "Card-based" interface. Related information is grouped within distinct containers featuring rounded corners and subtle backgrounds. This creates clear visual boundaties, making the data highly scannable for user.

3. **Core Screen Architecture**

   - **Dashboard Screem:**

     Utilizes a vertical layout focusing on a hero section for the total balance, followed by a summarizes list of recent transactions.

   - **History Screen:**

Employs a full-screen ***FlatList*** Layout for scrolling through extensive lists of data.

- **Add Transaction Screen:**
  Structed as a clean, highly legible data entry from with large touch targets amd floating input fields, ensure ease of use during one-handed operation.

- **Analytics Screen:**
  Features a specialized layout designed to allocate for data visualization components.

A.2 Use of Components

1. Reusability (Custom Components)

Reusability is use to avoid the rewriting same UI and the logic multiple times, that especially in inserts or updates workflows. In Personal Finance Tracker of the reusable custom component is shared to all the main screens, so that users can get the same navigation and the quick add behaviour when they are performing the transaction action. In the main reusable component is FinanceTabBar.tsx in the components/FinanceTabBar.tsx and plugged into the MainTabsNavigator once to serve all the main tabs.

In MainTabsNavigator.tsx of reuses one the custom tab bar for all pages:

```
<FinanceTabBar
  state={state as any}
  descriptors={{} as any}
  navigation={mockTabBarNavigation as any}
/>
```

The custom component is used:

FinanceTabBar.tsx used in MainTabsNavigator.tsx are shared across the four main pages: Dashboard, History, Analytics, Profile.

In this mean that the tab UI is implemented once and the reused everywhere instead of creating separate tab controls inside each of the screen.

The reusability supports inserts/updates:

Insert Flow:

The floating action button (+) inside the FinanceTabBar.tsx navigates to the AddTransaction.tsx giving users a consistent way to insert the new records from the main tab. This ensure that the "Insert" workflow is globally accessible without of the redundant button in the Dashboard, History or Analythics UI code.

In FinanceTabBar.tsx reusable insert entry point (+ button) to the AddTransaction.tsx:

```
<Pressable
  style={[styles.fab, { backgroundColor: colors.card }]}
  onPress={() => {
    const parentNav = navigation.getParent();
    if (parentNav) {
      const { width, height } = Dimensions.get('window');
      const fabCenterY = height - (Platform.OS === 'ios' ? 76 : 72);
      (parentNav as any).navigate('AddTransaction', {
        fromFab: true,
        originX: width / 2,
        originY: fabCenterY,
        fast: true, // Hint for snappier animation
      });
    }
  }}
>
  <View style={[styles.fabInner, { backgroundColor: colors.primary, shadowColor: colors.primary }]}>
    <Text style={styles.fabIcon}>+</Text>
  </View>
</Pressable>
```

Update Flow:

The same reusable tab bar that provides the consistent navigation to data related screens such as History and Profile where users that access and perform the update or edit behaviours.

Reusable of interaction behaviour:

AnimatedTabButton is reused for every tab item, so that all tabs can share the same press animation. This creation of the isolated functional component to abstracts the styling, press feedback and animated micro interactions such as scaling and translating on press. It used to every tab item, so that it never duplicates the animation logic.

In FinanceTabBar.tsx of the reusable tab button abstraction:

```
function AnimatedTabButton({
  isFocused,
  label,
  onPress,
  routeName,
  onLayout,
  activeColor,
  inactiveColor,
  indicatorBg,
}: {
  isFocused: boolean;
  label: string;
  onPress: () => void;
  routeName: string;
  onLayout?: (event: LayoutChangeEvent) => void;
  activeColor: string;
  inactiveColor: string;
  indicatorBg: string;
}) {
  const scale = useRef(new Animated.Value(1)).current;
  const translateY = useRef(new Animated.Value(0)).current;

  useEffect(() => {
    Animated.spring(translateY, {
      toValue: isFocused ? -4 : 0,
      useNativeDriver: true,
      friction: 8,
      tension: 100,
    }).start();
  }, [isFocused]);

  const handlePressIn = () => {
    Animated.spring(scale, {
      toValue: 0.9,
      useNativeDriver: true,
    }).start();
  };
```

```
const handlePressOut = () => {
  Animated.spring(scale, {
    toValue: 1,
    useNativeDriver: true,
  }).start();
};

return (
  <Pressable
    onPress={onPress}
    onPressIn={handlePressIn}
    onPressOut={handlePressOut}
    onLayout={onLayout}
    style={styles.tabButtonShell}
  >
    <Animated.View style={[styles.tabButton, { transform: [{ scale }, { translateY }] }]}>
      <View style={[styles.activeIndicatorBg, isFocused && styles.activeIndicatorBgVisible, { backgroundColor: indicatorBg }]} />
      <TabIcon routeName={routeName} color={isFocused ? activeColor : inactiveColor} />
      <Text numberOfLines={1} style={[styles.label, { color: isFocused ? activeColor : inactiveColor }]}>
        {label}
      </Text>
    </Animated.View>
  </Pressable>
);
```

Reusable tab metadata:

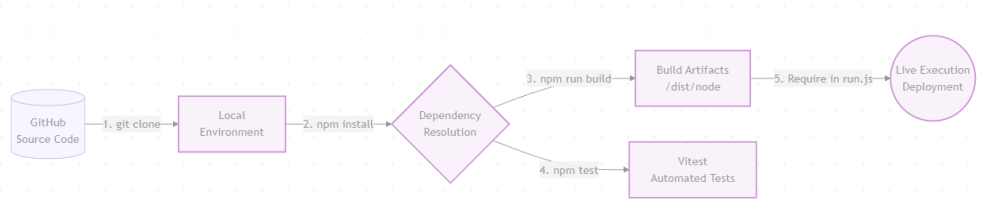
tabMeta is centralizes the tab labels in one location to reduce the hardcoded duplicates and make the future UI changes easier.

<u>In FinanceTabBar.tsx the reusable tab label mapping in one place:</u>

```
const tabMeta: Record<string, { label: string }> = {
  Dashboard: { label: 'Home' },
  History: { label: 'History' },
  Analytics: { label: 'Stats' },
  Profile: { label: 'Profile' },
};
```

2. Implementation of React-Navigation

To provide a seamless and scalable user experience, the application implements a sophisticated 2-tier nested navigation architecture utilizing the ***@react-navigation*** library. Rather than relying entirely on default navigator templates, we engineered a custom solution that combines a standard Stack Navigator with a highly customized Tab Navigator implementation. This structure ensures logical screen hierarchy, fluid swipe transitions, and centralized control over the application's routing state.



1. Root Stack Navigator

The foundation of our navigation is the ***RootStackNavigator,*** implemented using ***@react-navigation/native-stack.*** This acts as the primary router, handling top-level deep-linking, authentication flow and pushing full-screen modals.

For instance, pushing the ***AddTransactionScreen*** on top of the current view. This implements a Last-in-First-out (LIFO) stack, allowing users to intuitively navigate back to previous screen using the back button.

```
import { createNativeStackNavigator } from '@react-navigation/native-stack';
import MainTabsNavigator from './MainTabsNavigator';
import AddTransaction from '../screens/AddTransaction';
import LoginScreen from '../screens/LoginScreen';
```

```
export type RootStackParamList = {
  Login: { prefillEmail?: string; registeredName?: string } | undefined;
  Register: undefined;
  ForgotPassword: { prefillUsername?: string } | undefined;
  HelpSupport: undefined;
  PrivacyPolicy: undefined;
  MainTabs: undefined;
  AddTransaction: { fromFab?: boolean; originX?: number; originY?: number } | undefined;
```

```
return (
  isLoading ? (
    <View style={{ flex: 1, backgroundColor: '#090a1f', justifyContent: 'center', alignItems: 'center' }}>
      <ActivityIndicator size="large" color="#7f5bff" />
    </View>
  ) : (
    <Stack.Navigator
      initialRouteName={initialRoute}
      screenOptions={{
        headerTintColor: '#f5f7ff',
        headerStyle: { backgroundColor: '#090a1f' },
        headerShadowVisible: false,
        contentStyle: { backgroundColor: '#090a1f' },
      }}
    >
      <Stack.Screen name="Login" component={LoginScreen} options={{ headerShown: false }} />
```

```
<Stack.Screen
  name="MainTabs"
  component={MainTabsNavigator}
  options={{ headerShown: false }}
/>
<Stack.Screen
  name="AddTransaction"
  component={AddTransaction}
  options={{
    headerShown: false,
    animation: 'none',
    presentation: 'transparentModal',
    contentStyle: { backgroundColor: 'transparent' }
  }}
```

2. Custom Tab Navigator & PageView

Instead of using the standard ***@react-navigation/bottom-tabs***, we implement a highly customized ***MainTabNavigator***. To achive smooth, native-feeling swipe gestures between main screens, we utilized ***react-native-pager-view***.

This PagerView is then synchronized with our custom reusable component, ***FinanceTabBar*** hooks, we created a ***PageWrapper*** bridge. This custom implementation prevents UI locking during heavy data renders and allows tabs to share a single animated interaction logic

```
import React, { useRef, useState, useMemo, useEffect, createContext, useContext } from 'react';
import { View, StyleSheet } from 'react-native';
import PagerView from 'react-native-pager-view';
import Dashboard from '../screens/Dashboard';
import History from '../screens/History';
import Analytics from '../screens/Analytics';
import Profile from '../screens/Profile';
import FinanceTabBar from '../components/FinanceTabBar';
```

```
export default function MainTabsNavigator({ navigation }: any) {
  const [index, setIndex] = useState(0);
  const pagerRef = useRef<PagerView>(null);

  const routes = useMemo(() => [
    { name: 'Dashboard', key: 'Dashboard' },
    { name: 'History', key: 'History' },
    { name: 'Analytics', key: 'Analytics' },
    { name: 'Profile', key: 'Profile' },
  ], []);
```

```
return (
  <PagerContext.Provider value={{ jumpTo }}>
    <View style={styles.container}>
      <PagerView
        ref={pagerRef}
        style={styles.pager}
        initialPage={0}
        onPageSelected={(e) => {
          const newIndex = e.nativeEvent.position;
          setIndex(newIndex);
        }}
```

```
  <View key="Dashboard" style={styles.page}>
    <PageWrapper
      component={Dashboard}
      isFocused={index === 0}
      navigation={navigation}
      route={routes[0]}
    />
  </View>
```

```
      <FinanceTabBar
        state={state as any}
        descriptors={{} as any}
        navigation={mockTabBarNavigation as any}
      />
    </View>
</PagerContext.Provider>
```

Part B

DATA PERSISTENCE

B.1 Usage of Data Persistence

The Personal Finance Tracker app relies on data persistence as a foundational mechanism to ensure that all financial information remains accessible across app sessions, device restarts, and server reboots.

The following types of data are persisted:

- Transactions: records of income and expenses linked to a wallet and category.
- Users: account information including credentials for authentication.
- Wallets: the financial accounts that belong to each user.
- Budgets: spending limits configured per category and time period.
- Recurring transactions: rules that automatically generate transactions on a set interval.
- Authentication sessions: tokens that keep a user securely logged in.

The app also handles sensitive user account information and authentication sessions. To protect this sensitive data and align with industry-standard practices, user authentication state is persisted through dedicated sessions and users tables, and passwords are encrypted using the scrypt hashing algorithm prior to storage. This ensures that plaintext credentials are never written to disk.

To manage this data reliably, the app employs a two-tier persistence architecture: a server-side SQLite database for authoritative data management and a client-side SQLite store for offline functionality.

1. Server-Side Persistent Storage

On the server side, the primary persistent store is a file-based SQLite database located at finance_tracker.sqlite. Managed by the backend service (service.js), this database acts as the single source of truth for all authoritative reads and

writes. Because it is file-based, data remains intact even during server reboots. Furthermore, this backend architecture is designed with robust operational and maintenance considerations. Schema migrations and structural alterations—such as adding new columns—are performed automatically at server startup through ensureTables and column-check routines, ensuring backward compatibility as new features are introduced. For debugging and auditing during development, the database file can also be externally backed up and inspected using provided tooling (inspect_db.js).

2. Client-Side Persistent Storage

On the mobile client, the application utilizes react-native-sqlite-storage (implemented via sqlite.ts) for local on-device persistence of transaction records. This client-side relational database is essential for enabling full offline CRUD operations, allowing users to seamlessly record and browse transactions without an active network connection. Once network connectivity is restored, the mobile app synchronises with the server-side endpoints, ensuring the backend database remains the definitive source of truth across multiple devices. In development or testing environments where the native SQLite module is unavailable, the client gracefully falls back to an in-memory store to keep the app functional. Finally, while @react-native-async-storage/async-storage is available within the project, its role is strictly limited to lightweight key-value caching and basic application settings, reserving the robust SQLite implementation entirely for primary transactional data.

B.2 Implementation of Data Persistence

The implementation of data persistence in the Personal Finance Tracker employs a structured, multi-layered approach that ensures both data durability and application reliability. This section details how SQLite persistence is integrated, configured, and maintained across the server and mobile client.

1. Server-Side Implementation

The backend service, managed by service.js, initializes and maintains the SQLite database through a series of setup routines executed at server startup. The ‘ensureTables()’ function programmatically creates the required database schema if it does not already exist, including six core tables: ‘users’, ‘sessions’, ‘password_resets’, ‘wallets’, ‘budgets’, ‘transactions’, and ‘recurring_transactions’. Each table is defined with strict schema constraints: primary keys for uniqueness, foreign-key relationships for referential integrity, and CHECK constraints to enforce valid data types (for example, the ‘type’ column in transactions is restricted to ‘income’, ‘expense’, or ‘transfer’). Additionally, indexes are created on frequently queried columns, such as ‘idx_transactions_user_date’, to optimize retrieval performance for large datasets.

```
function ensureTables(onDone) {
  const db = openDb();
  db.serialize(() => {
    db.run('PRAGMA foreign_keys = ON');

    // Auth tables
    db.run(`
      CREATE TABLE IF NOT EXISTS users (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        username TEXT NOT NULL UNIQUE,
        password TEXT NOT NULL,
        email TEXT,
        phone TEXT,
        profilePic TEXT,
        createdAt TEXT NOT NULL
      )
    `);
```

```
db.run(`
  CREATE INDEX IF NOT EXISTS idx_transactions_user_date
  ON transactions(userId, date DESC)
`);
```

To maintain backward compatibility as the application evolves, the server performs automated schema migrations at startup through column-check routines. For instance, the ‘ensureTransactionColumns()’ function inspects the existing schema using PRAGMA directives and dynamically adds new columns

(such as ‘note’ or ‘receiptUrl’) if they are missing, allowing existing databases to upgrade gracefully without manual intervention or data loss.

```
function ensureTransactionColumns(db, onDone) {
  db.all('PRAGMA table_info(transactions)', [], (err, rows) => {
    if (err) {
      console.error(err.message);
      if (typeof onDone === 'function') {
        onDone();
      }
      return;
    }

    const existingColumns = new Set((rows || []).map(column => column.name));
    const alterStatements = [];
```

Data integrity is enforced through parameterized SQL queries throughout the backend. All user inputs—whether from authentication requests or transaction creation—are bound as query parameters rather than concatenated into SQL strings. This practice prevents SQL injection vulnerabilities and ensures that data is properly escaped and typed. For example, transaction creation uses prepared statements with parameter placeholders (‘?’), and all inputs undergo validation before database insertion.

```
// Transaction Endpoints
app.get('/api/transactions', requireAuth, (req, res) => {
  const { walletId } = req.query;
  const db = openDb();
  let sql = 'SELECT * FROM transactions WHERE userId = ?';
  let params = [req.auth.userId];

  if (walletId) {
    sql += ' AND walletId = ?';
    params.push(walletId);
  }

  sql += ' ORDER BY date DESC';
```

Password security is implemented using the scrypt key-derivation function. When a user registers or changes their password, the plaintext password is hashed with a randomly generated salt, and only the salt-hash pair is stored in the database. Authentication is verified by re-hashing the provided password with the stored salt and comparing the result using a timing-safe comparison function to prevent timing attacks.

```
function hashPassword(password) {
  const salt = crypto.randomBytes(16).toString('hex');
  const hash = crypto.scryptSync(password, salt, 64).toString('hex');
  return `${salt}:${hash}`;
}

function verifyPassword(password, stored) {
  const parts = String(stored || '').split(':');
  if (parts.length !== 2) {
    return stored === password;
  }
```

Authentication state is managed through the ‘sessions’ table, which stores login tokens linked to user IDs and creation timestamps. Tokens are cryptographically generated using ‘crypto.randomBytes()’ to ensure uniqueness and unpredictability. Upon login, a new session record is created; upon logout, the corresponding record is deleted. The backend validates every authenticated request by querying the sessions table to verify the provided bearer token.

```
function generateToken() {
  return crypto.randomBytes(24).toString('hex');
}
```

```
function requireAuth(req, res, next) {
  const token = getBearerToken(req);
  if (!token) {
    return res.status(401).json({ message: 'Missing auth token' });
  }

  const db = openDb();
  db.get(
    `SELECT sessions.userId, users.username
     FROM sessions
     JOIN users ON users.id = sessions.userId
     WHERE sessions.token = ?`,
    [token],
    (err, row) => {
      closeDb(db);

      if (err) {
        return res.status(500).json({ message: 'Database error' });
      }

      if (!row) {
        return res.status(401).json({ message: 'Invalid or expired token' });
      }

      req.auth = {
        token,
        userId: Number(row.userId), // Store as Number for consistent DB querying
        username: row.username,
      };
      return next();
    }
  );
}
```

2. Client-Side Implementation

On the mobile client, persistent storage is implemented through sqlite.ts, which wraps the 'react-native-sqlite-storage' module. The client initializes the database by calling 'openDatabase()', which establishes a connection to a named database stored on the device's local file system. The schema is similar to the server schema but focused on the transactions table, allowing users to store transaction records locally.

```
const isJestRuntime = typeof process !== 'undefined' && Boolean(process.env.JEST_WORKER_ID);
const hasSQLiteModule = Boolean(SQLiteModule?.openDatabase) && !isJestRuntime;
let sqliteNativeUnavailable = false;
```

```
async function openDatabase(): Promise<any> {
  if (!hasSQLiteModule || sqliteNativeUnavailable) {
    return null;
  }

  if (!dbConnectionPromise) {
    try {
      dbConnectionPromise = SQLiteModule.openDatabase({
        name: config.sqliteDbName,
        location: 'default',
      });
    } catch {
      sqliteNativeUnavailable = true;
      dbConnectionPromise = null;
      return null;
    }
  }

  return dbConnectionPromise;
}
```

The client module implements a graceful fallback mechanism: if the native SQLite module is unavailable (for example, during Jest testing or if native module linking fails), the application falls back to an in-memory array-based store. This fallback is transparent to the calling code and allows the application to remain functional across different environments without code modification.

```
try {
  // Use runtime require to avoid breaking Jest (native module unavailable there).
  // eslint-disable-next-line @typescript-eslint/no-var-requires
  SQLiteModule = require('react-native-sqlite-storage');
  if (typeof SQLiteModule?.enablePromise === 'function') {
    SQLiteModule.enablePromise(true);
  }
} catch {
  SQLiteModule = null;
}
```

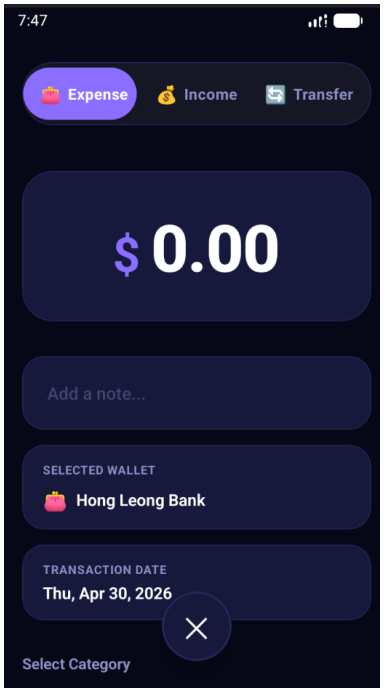
Data initialization on the client is handled by 'initDatabase()', which ensures the transactions table exists and populates it with seed data if configured. All

client-side CRUD operations are performed asynchronously using promises, ensuring that long-running database operations do not block the user interface.

```
export async function insertTransaction(input: CreateTransactionInput): Promise<Transaction> {
  const normalized = toDbTransaction(input);
  const newRow: Transaction = {
    ...normalized,
    id: nextId(),
  };

  if (!hasSQLiteModule || sqliteNativeUnavailable) {
    memoryStore = sortTransactions([newRow, ...memoryStore]);
    await emitChange();
    return newRow;
  }

  await initDatabase();
  await executeSql(
    `INSERT INTO ${TABLE_NAME} (id, amount, type, category, date, note, receiptUrl, userId)
     VALUES (?, ?, ?, ?, ?, ?, ?, ?)`,
    [
      newRow.id,
      newRow.amount,
      newRow.type,
      newRow.category,
      newRow.date,
      newRow.note ?? '',
      newRow.receiptUrl ?? '',
      newRow.userId,
    ],
  );

  await emitChange();
  return newRow;
}
```

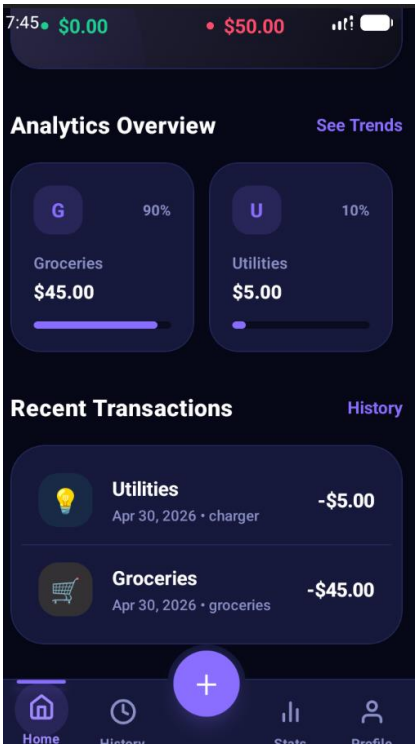
Input validation and data normalization occur before insertion. For instance, the ‘toDbTransaction()’ function normalizes transaction amounts to numbers, ensures optional fields (like notes and receipt URLs) have default values, and applies consistent formatting to prepare data for reliable storage.

```
function toDbTransaction(input: CreateTransactionInput): CreateTransactionInput {
  return {
    ...input,
    amount: Number(input.amount),
    note: input.note ?? '',
    receiptUrl: input.receiptUrl ?? '',
  };
}
```



Offline-first architecture is supported through the client SQLite store: users can create, read, update, and delete transactions without a network connection. The client maintains a list of listeners (observer pattern) that are notified whenever data changes, allowing the UI to reflect updates in real time. When network connectivity is restored, a synchronization mechanism sends pending changes to the server, which acts as the authoritative source.

```
const seedRows: Transaction[] = [
  createSeedTransaction('seed-1', 4200, 'income', 'salary', '2026-04-04T08:30:00Z', 'Monthly salary'),
  createSeedTransaction('seed-2', 84.2, 'expense', 'groceries', '2026-04-04T11:45:00Z', 'Weekend grocery run'),
  createSeedTransaction('seed-3', 14.9, 'expense', 'transport', '2026-04-03T15:15:00Z', 'Grab ride'),
  createSeedTransaction('seed-4', 120.0, 'expense', 'utilities', '2026-04-02T09:00:00Z', 'Water bill'),
  createSeedTransaction('seed-5', 250.0, 'income', 'freelance', '2026-04-01T19:00:00Z', 'Side project payment'),
];

let memoryStore: Transaction[] = [...seedRows];
```



3. Error Handling and Robustness

Both the server and client implement error handling to manage database failures gracefully. On the server, if a database operation fails, the error is logged and an appropriate HTTP status code is returned to the client (for example, 500 for internal errors, 400 for validation failures). On the client, if SQLite operations fail, the application falls back to the in-memory store and continues functioning, ensuring that temporary database issues do not crash the application.

The implementation also includes audit and debugging features. The inspect_db.js utility provides developers with a command-line interface to query the server database directly, inspect table schemas, and verify data integrity during development and troubleshooting.

```
const sqlite3 = require('sqlite3').verbose();
const db = new sqlite3.Database('./db/finance_tracker.sqlite');

db.all("SELECT name, sql FROM sqlite_master WHERE type='table'", (err,
  if (err) {
    console.error(err);
    process.exit(1);
  }

  tables.forEach(table => {
    console.log(`\nTable: ${table.name}`);
    console.log('SQL:', table.sql);

    db.all(`PRAGMA table_info(${table.name})`, (err, columns) => {
      console.log('Columns:', columns);
    });
  });
```

Through this comprehensive implementation, the Personal Finance Tracker achieves reliable, secure, and user-friendly data persistence that supports both online and offline workflows while maintaining data integrity and security.

B.3 CRUD Operations & Input Validation

B.3.1 Create

The create operation is implemented with using the offline-first approach to ensure the seamless user experience even without the active internet connection. In the backend section, the POST routes designed for creation such as the transaction endpoint in service.js that are protected by the shared requireAuth middleware. The server validates the incoming payload that to ensure the required fields such as the amount, type and category are present and correct, then execute the parameterized SQL queries to secure the insert the new record into the database that linked with the authenticated user's ID.

In service.js:

```
app.post('/api/transactions', requireAuth, (req, res) => {
  const { amount, type, category, date, note, receiptUrl, walletId } = req.body || {};
  if (!amount || !type || !category || !date) return res.status(400).json({ message: 'Missing fields' });

  const db = openDb();
  db.run(
    `INSERT INTO transactions(userId, amount, type, category, note, date, createdAt, receiptUrl, walletId)
     VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)`,
    [req.auth.userId, Number(amount), type, category, note || '', date, new Date().toISOString(), receiptUrl || '', walletId || null],
    function(err) {
      closeDb(db);
      if (err) return res.status(500).json({ message: 'Database error' });
      emitBudgetAlertsForUser(req.auth.userId, date);
      res.status(201).json({ id: this.lastID, affected: this.changes });
    }
  );
});
```

In the client side, the creation flow is managed by local database wrappers and API services. When a user that submits a new transaction, the app immediate writes the record into the database. At the same time, it queues the action and

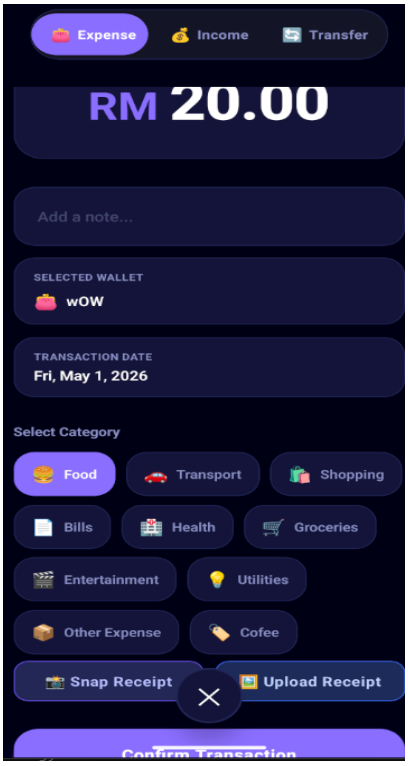
the triggers a background of the synchronization process of syncPendingTransactions(). This guarantees that the UI is updates instantly with the new data, the actual of REST API request to the server is reliably assign in the background as soon as the network connectivity is available.

In transactionApi.ts:

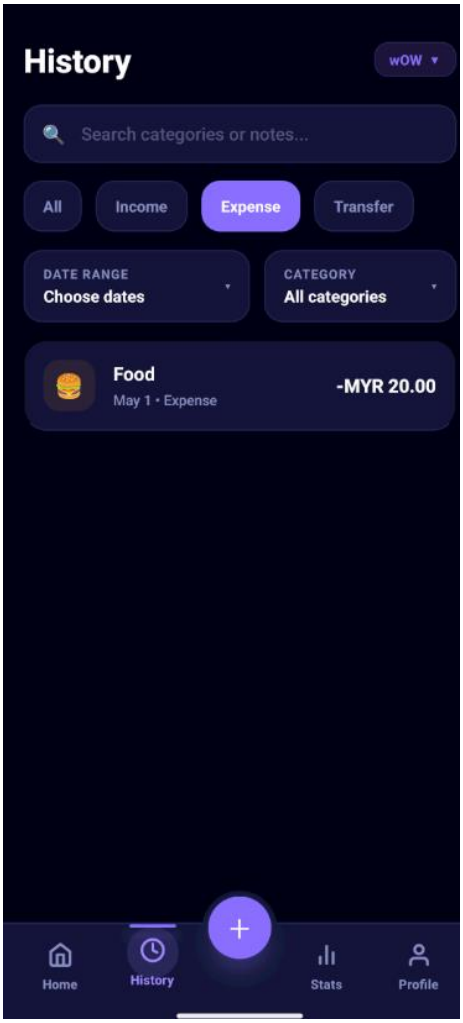
```
export async function insertTransaction(input: CreateTransactionInput): Promise<Transaction> {
  const session = getAuthSession();
  const localTransaction = await insertTransactionLocal({
    ...input,
    userId: session?.userId ?? input.userId,
  });

  await addPendingTransactionChange('create', localTransaction.id, localTransaction);
  setPendingCount((await getPendingTransactionChanges()).length);
  clearTransactionCache();
  void syncPendingTransactions();
  return localTransaction;
}
```

syncPendingTransactions():

```
export async function syncPendingTransactions(): Promise<void> {
  if (isSyncInProgress) {
    return;
  }
```



After Confirm Transaction:



B.3.2 Read

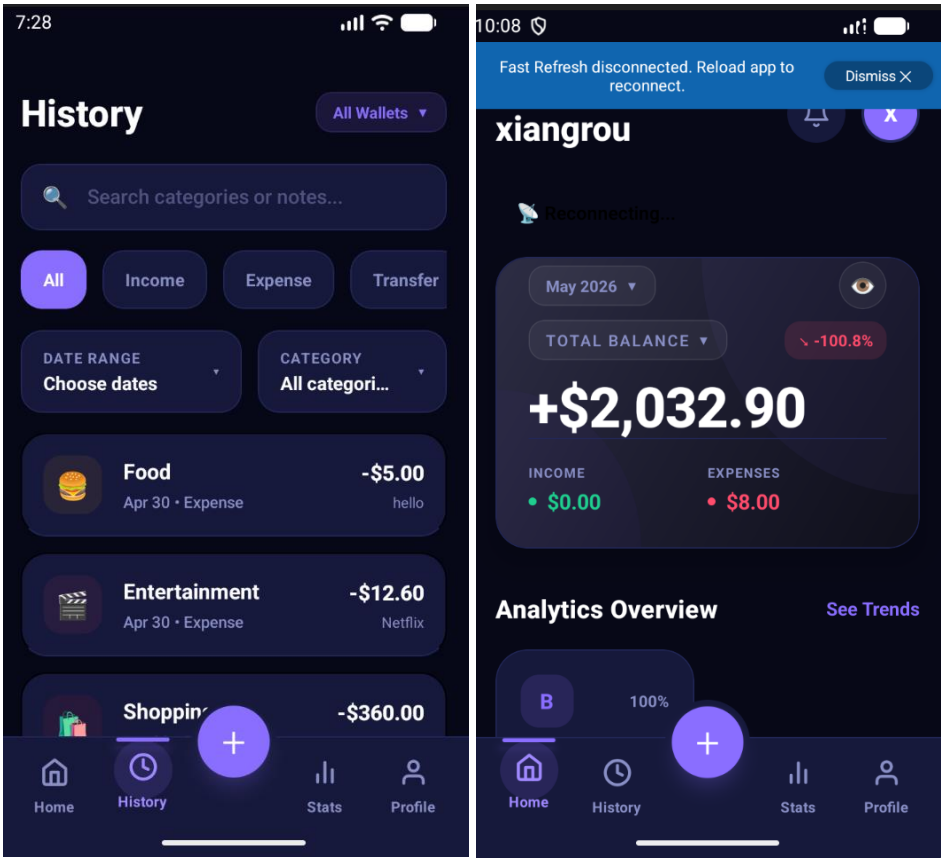
The application uses a read-first CRUD flow for presenting financial data. On the backend, every read route is protected by authentication through the shared requireAuth middleware, so only the signed-in user can retrieve their own records. The main transaction read endpoint returns the user's transactions and supports an optional walletId filter. Similar read-side validation is applied to budget and analytics requests. Both require a month query parameter and reject the request if it is missing. The auth gate used by these routes is defined in service.js.

```
// Transaction Endpoints
app.get('/api/transactions', requireAuth, (req, res) => {
  const { walletId } = req.query;
  const db = openDb();
  let sql = 'SELECT * FROM transactions WHERE userId = ?';
  let params = [req.auth.userId];

  if (walletId) {
    sql += ' AND walletId = ?';
    params.push(walletId);
  }

  sql += ' ORDER BY date DESC';

  db.all(sql, params, (err, rows) => {
    closeDb(db);
    if (err) return res.status(500).json({ message: 'Database error' });
    res.status(200).json(rows || []);
  });
});
```

```
// Budget Endpoints
app.get('/api/budgets', requireAuth, (req, res) => {
  const month = (req.query.month || '').toString();
  if (!month) return res.status(400).json({ message: 'month query is required, format YYYY-MM' });

  const db = openDb();
  db.all(
    'SELECT id, category, month, amount, createdAt FROM budgets WHERE userId = ? AND month = ? ORDER BY category ASC',
    [req.auth.userId, month],
    (err, rows) => {
      closeDb(db);
      if (err) return res.status(500).json({ message: 'Database error' });
      res.status(200).json(rows || []);
    }
  );
});
```

On the client side, the read path is handled by transactionApi.ts. It first loads local SQLite data for offline access, then fetches the server version and merges both sources so the UI stays stable even when the network is slow or unavailable. If the server request fails, the app falls back to the local data instead of showing an error state. This makes the read operations reliable and ensures the dashboard and history screens can still display financial records when offline.

```
const fallbackUserId = getUserIdForFallback(_userId);
const localRows = fallbackUserId ? await getTransactionsByUserLocal(fallbackUserId) : await getAllTransactions();
const localData = walletId
  ? localRows.filter(item => String(item.walletId ?? '') === String(walletId))
  : localRows;

const pendingChanges = await getPendingTransactionChanges();

await syncPendingTransactions();

let url = `${config.apiBaseUrl}/api/transactions`;
if (walletId) {
  url += `?walletId=${encodeURIComponent(walletId)}`;
}

try {
  const response = await fetch(url, {
    headers: {
      ...getAuthHeaders(),
    },
  });

  const rows = await parseResponse<any[]>(response);
  const serverData = rows.map(mapTransactionRow);
  const filteredServerData = walletId
    ? serverData.filter(item => String(item.walletId ?? '') === String(walletId))
    : serverData;
```



3. Update

The update opearation follow the same delayed, offline side architecture as the create and delete opearation to guarantee the strong user experience accross changes of network conditions.

On the backend, the PUT endpoint in service.js is secured by the requireAuth middleware, ensure the user can only edit their own transaction records. The server utilizes a parameterized UPDATE SQL statement to sanitize the input such as the amount, category and note thereby protecting against the SQL injection vulnerabilities. Once the database is successfully updated, the server recalculates the limit of metrices by invocking emitBudgetAlertsForUser() before returning to the success status.

In service.js:

```
app.put('/api/transactions/:id', requireAuth, (req, res) => {
  const { amount, type, category, date, note, receiptUrl, walletId } = req.body || {};
  const db = openDb();
  db.run(
    `UPDATE transactions SET amount = ?, type = ?, category = ?, date = ?, note = ?, receiptUrl = ?, walletId = ?
     WHERE id = ? AND userId = ?`,
    [Number(amount), type, category, date, note || '', receiptUrl || '', walletId || null, req.params.id, req.auth.userId],
    function(err) {
      closeDb(db);
      if (err) return res.status(500).json({ message: 'Database error' });
      emitBudgetAlertsForUser(req.auth.userId, date);
      res.status(200).json({ ok: true, affected: this.changes });
    }
  );
});
```

EmitBudgetAlertsForUser():

```
function emitBudgetAlertsForUser(userId, dateValue) {
  if (!financeNamespace) {
    return;
  }

  const targetDate = dateValue ? new Date(dateValue) : new Date();
  if (Number.isNaN(targetDate.getTime())) {
    return;
  }

  const monthKey = targetDate.toISOString().slice(0, 7);
  const db = openDb();

  db.all(
    `SELECT b.category, b.amount AS budgetAmount,
            COALESCE(SUM(CASE WHEN t.type = 'expense' THEN t.amount ELSE 0 END), 0) AS actualAmount
     FROM budgets b
     LEFT JOIN transactions t
       ON t.userId = b.userId
      AND t.category = b.category
      AND substr(t.date, 1, 7) = b.month
     WHERE b.userId = ? AND b.month = ?
     GROUP BY b.category, b.amount`,
    [userId, monthKey],
    (err, rows) => {
      closeDb(db);

      if (err || !rows || rows.length === 0) {
        return;
      }

      rows.forEach(row => {
        const budgetAmount = Number(row.budgetAmount) || 0;
        const actualAmount = Number(row.actualAmount) || 0;
        if (budgetAmount > 0 && actualAmount >= budgetAmount * 0.9) {
          financeNamespace.to(`user_${userId}`).emit('budget_alert', {
            category: row.category,
            spent: Math.round(actualAmount * 100) / 100,
            limit: Math.round(budgetAmount * 100) / 100,
            percentUsed: Math.round((actualAmount / budgetAmount) * 100),
            message: `${row.category} budget is ${Math.round((actualAmount / budgetAmount) * 100)}% used for ${monthKey}`,
          });
        }
      });
    }
  );
}
```

In client side, the update flow inside transactionApi.ts modifies the target row instantly in the local SQLite database(updateTransactionLocal()) so that the UI updates immediately. Instead of waiting the successfull HTTP request, it logs an ‘update’ action along with the newly modified payload in the pending_transaction_changes table via addPendingTransactionChange(). The API cache is then cleared to seamlessly refresh the dashboard readings, the

syncPendingTransaction() is invoked in the background. When the internet connection is available, the sync layer loops over the pending queue and the reliably pushed the updated data to the server.

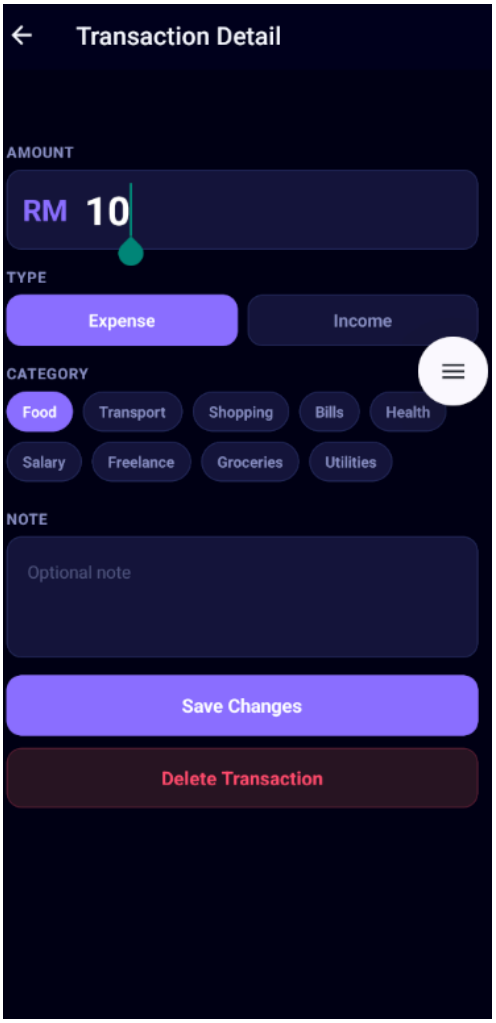
In transactionApi.ts:

```
export async function updateTransaction(
  id: string,
  updates: TransactionUpdate,
  fallback?: Transaction,
): Promise<Transaction | null> {
  const next = {
    amount: Number(updates.amount ?? fallback?.amount ?? 0),
    type: updates.type ?? fallback?.type,
    category: updates.category ?? fallback?.category,
    date: updates.date ?? fallback?.date,
    note: updates.note ?? fallback?.note ?? '',
    receiptUrl: updates.receiptUrl ?? fallback?.receiptUrl ?? '',
    walletId: updates.walletId ?? fallback?.walletId,
  };

  if (!next.type || !next.category || !next.date || next.amount <= 0) {
    throw new Error('amount, type, category and date are required');
  }

  const result = await updateTransactionLocal(id, updates);
  if (!result) {
    return null;
  }

  await addPendingTransactionChange('update', id, {
    amount: result.amount,
    type: result.type,
    category: result.category,
    date: result.date,
    note: result.note ?? '',
    receiptUrl: result.receiptUrl ?? '',
    walletId: result.walletId,
  });
```

```
  clearTransactionCache();
  setPendingCount((await getPendingTransactionChanges()).length);
  void syncPendingTransactions();
  return result;
}
```

Before Update Transactions:

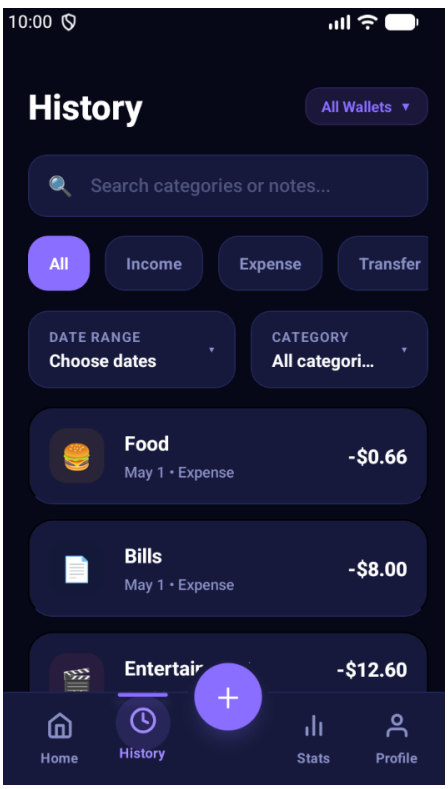
After Update:

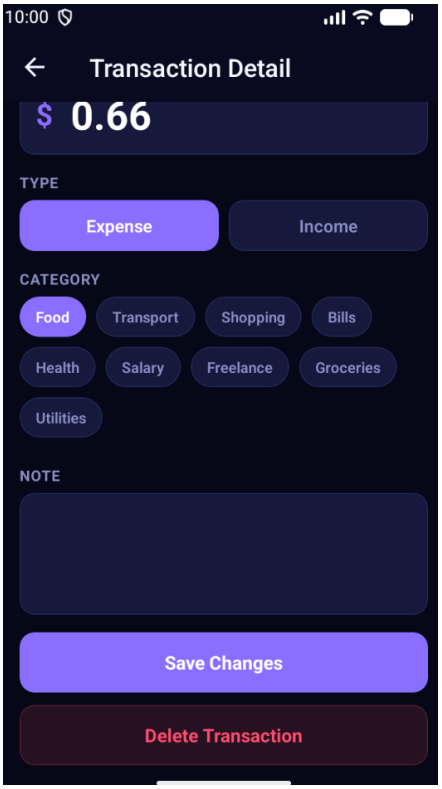

Transaction Detail
AMOUNT
RM 10
TYPE
Expense
Income
CATEGORY
Food
Transport
Shopping
Bills
Health
Salary
Freelance
Groceries
Utilities
NOTE
Optional note
Save Changes
Delete Transaction

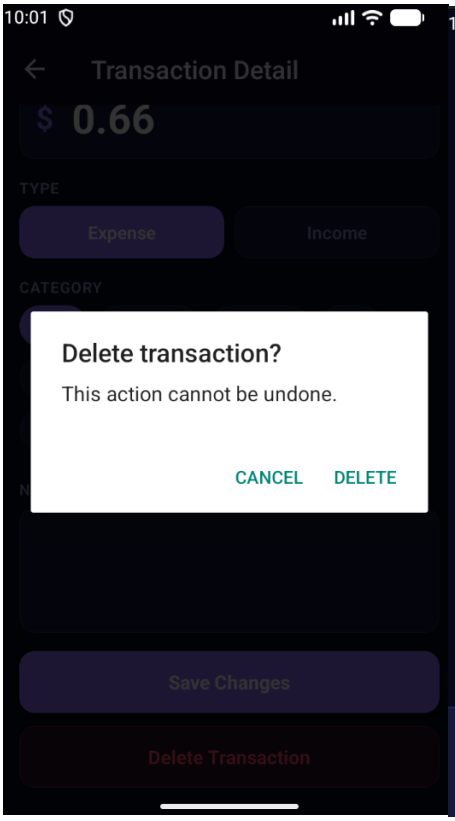
B.3.3 Delete

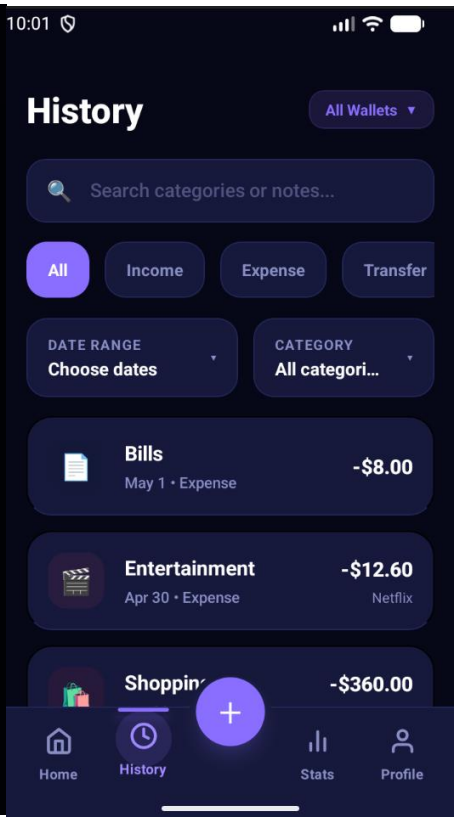
The delete operation is implemented as a deferred, offline-safe workflow. On the server, authenticated DELETE routes in service.js remove only the transaction that belongs to the current user.

On the client, transactionApi.ts does not delete the row from SQLite immediately. Instead, deleteTransaction() records a pending delete in pending_transaction_changes, clears the transaction cache, and triggers syncPendingTransactions(). When connectivity is available, the sync layer sends the DELETE request to the backend; after the server confirms success, the client removes the local SQLite row and clears the pending change. This approach preserves data consistency while still supporting offline use.


10:00
History
All Wallets
Search categories or notes...
All
Income
Expense
Transfer
DATE RANGE
Choose dates
CATEGORY
All categori...
Food
May 1 • Expense
-$0.66
Bills
May 1 • Expense
-$8.00
-$12.60
Home
History
Stats
Profile


10:00
Transaction Detail
$ 0.66
TYPE
Expense
Income
CATEGORY
Food
Transport
Shopping
Bills
Health
Salary
Freelance
Groceries
Utilities
NOTE
Save Changes
Delete Transaction


10:01
Transaction Detail
$ 0.66
TYPE
Expense
Income
CATEGORY
Delete transaction?
This action cannot be undone.
CANCEL
DELETE
Save Changes
Delete Transaction


10:01
History
All Wallets
Search categories or notes...
All
Income
Expense
Transfer
DATE RANGE
Choose dates
CATEGORY
All categori...
Bills
May 1 • Expense
-$8.00
Entertainment
Apr 30 • Expense
Netflix
-$12.60
-$360.00
Home
History
Stats
Profile

Part C

CLOUD CONNECTIVITY

C.1 Usage of Cloud Connectivity

The Personal Finance Tracker application heavily relies on cloud connectivity to ensure data synchronization across devices, deliver real-time notifications, and provide accurate, up-to-date financial contexts. The cloud connectivity features implemented in this application can be categorized into three primary usages:

1. **REST API Data Synchronization:** The application uses a robust RESTful API architecture to synchronize local data (transactions, wallets, budgets, and user profiles) with the central backend server. This allows users to access their financial data seamlessly across multiple devices while maintaining a single source of truth in the cloud.
2. **Real-Time WebSockets & Interactive Notifications:** To provide an interactive and responsive user experience, the application utilizes persistent WebSockets. This connection is crucial for delivering instant alerts, such as when a user's spending exceeds their allocated budget threshold, or when background recurring transactions are automatically synced. The application utilizes a highly decoupled Observer Pattern to broadcast these updates to the React UI in real time, persisting alerts into a dedicated notification centre.
3. **External Currency System Integration:** To cater to international financial tracking, the app connects to third-party financial APIs (Alpha Vantage) to fetch real-time currency exchange rates. This allows the application to dynamically convert and display transaction amounts between USD and MYR based on live market data, rather than relying on static, outdated rates.

C.1.1 REST API Data Synchronization:

Purpose and Benefits

The REST API synchronization layer ensures that user financial data (transactions, wallets, budgets, and profiles) remains consistent across multiple devices while maintaining a reliable offline-first experience. Users can:

- Create, Read, Update, and Delete transactions seamlessly online, Auto synchronization for offline new transactions
- Access financial data from multiple devices with automatic synchronization
- Work offline without losing any data which the pending changes are queued locally
- Maintain data integrity through a smart merge strategy that respects server-as-source-of-truth while protecting local changes

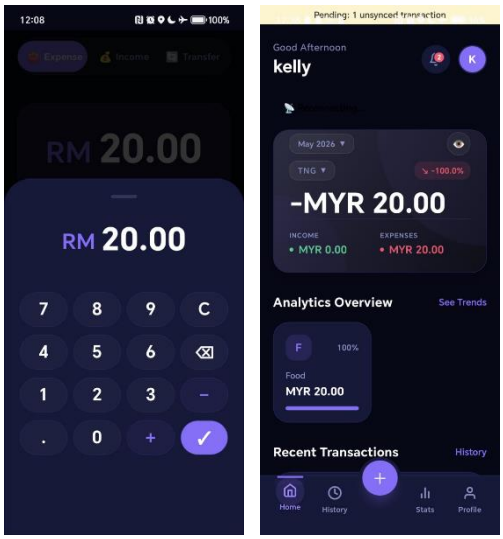
Key Features

Feature 1: Automatic Offline Support

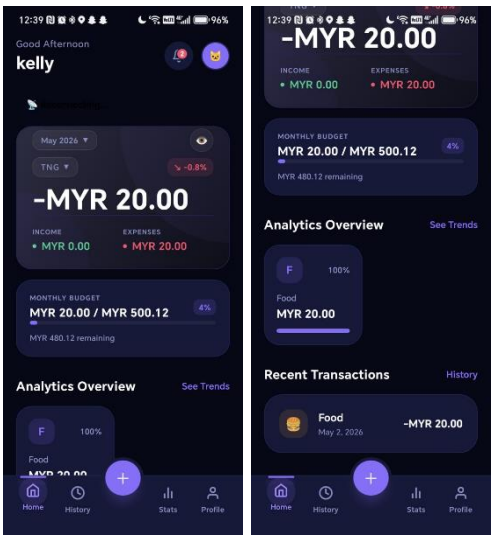
Users can add transactions even without internet connectivity. The application:

- Stores changes locally in SQLite database
- Queues pending operations in a dedicated change tracking table
- Automatically syncs when connectivity is restored

Add Transaction during Offline (Fly mode):



After Online (Auto-sync):



Feature 2: Multi-Device Synchronization

When a user logs in on a new device:

- Local transaction history is preserved
- Pending changes are synced first (maintains operation order)
- Server data is merged with local data
- Conflicts are resolved intelligently (local changes take priority for non-synced items)

Feature 3: Continuous Background Sync

The application runs a background sync process every 15 seconds:

- Monitors pending transaction changes
- Prioritizes older changes first (FIFO order)
- Updates pending count in UI for user awareness
- Stops if network errors occur to preserve operation order

C.1.2 Real-Time WebSocket Communication

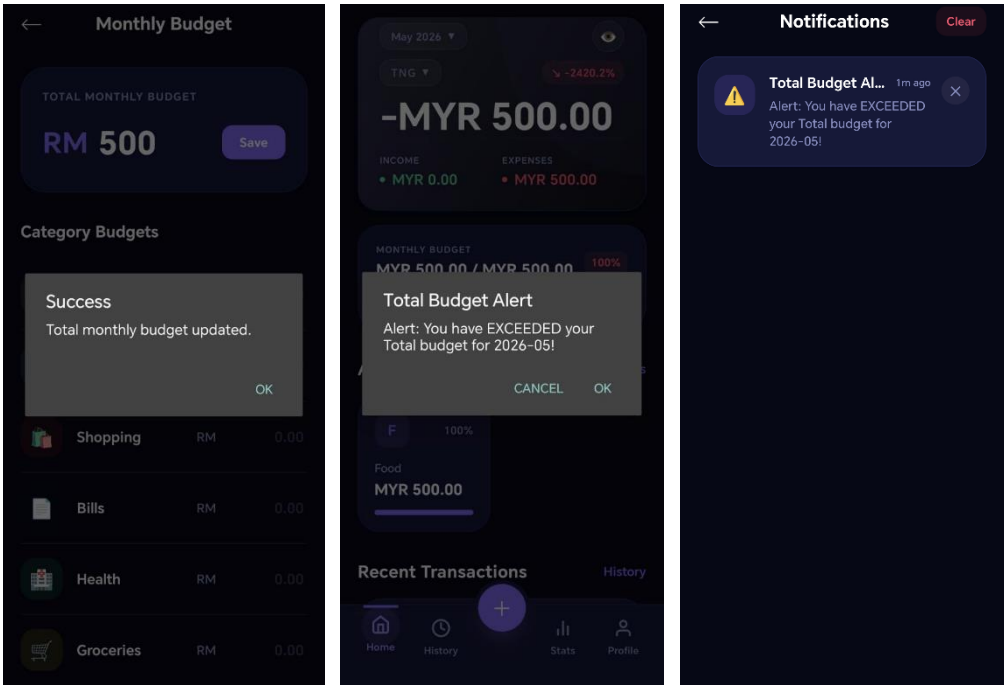
Purpose and Benefits

WebSockets provide instant, bidirectional communication between client and server, which have feature such as:

Feature 1: Budget Alert Notifications

When a user's spending in a category exceeds their budget which being set at setting:

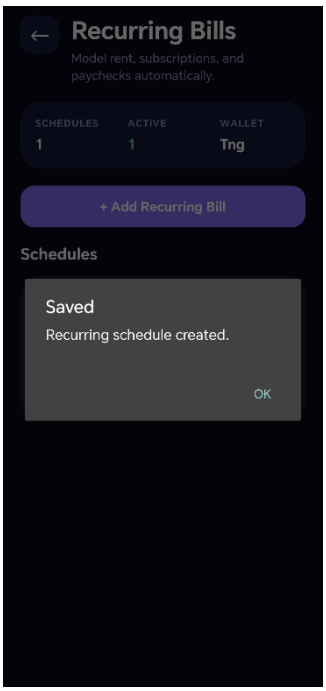
- Instant Alert: User receives immediate notification via WebSocket
- Actionable Information: Message helps user understand spending status immediately



Feature 2: Recurring Transaction Sync Notifications

When background recurring transactions are automatically synchronized:

- Sync Confirmation: Server notifies client of successful sync
- Count Information: User sees how many transactions were synced
- Persistent Storage: Notifications persist in notification centre



Feature 3: Connection Status Awareness

Application monitors WebSocket connection status:

- Displays online/offline indicator to user

```
C:\Users\Kelly\WAD_Assignment\Personal-Finance-Tracker\db>node service.js
🚀 Starting backend...
✓ Database tables created
✓ Database seeded
Server running on port 5001
Database: C:\Users\Kelly\WAD_Assignment\Personal-Finance-Tracker\db\finance_track
er.sqlite
✓ WebSocket listening on /finance namespace
✓ Backend ready for connections
[Socket] User connected: 74CiKEq4T9wJxtbcAAAB
[Socket] User 3 joined room
```

```
[Socket] User disconnected: mIIx7qTOWXMmWPdPAAAF
[Socket] User connected: a36mN2cEYb2uMGtaAAAH
[Socket] User 3 joined room
```

C.1.3 External Currency System Integration

Purpose and Benefits

The application integrates with Alpha Vantage API to provide real-time USD to MYR currency conversion:

Feature 1: Real-Time Exchange Rates

- Live Market Data: Exchange rates updated every 700s, 11.6 minutes from Alpha Vantage
- Automatic Conversion: Transactions automatically display in user's preferred currency
- Cached Rates: Falls back to cached rates during API outages
- Transparent Display: Users see both original amount and converted amount

Feature 2: Multi-Currency Support

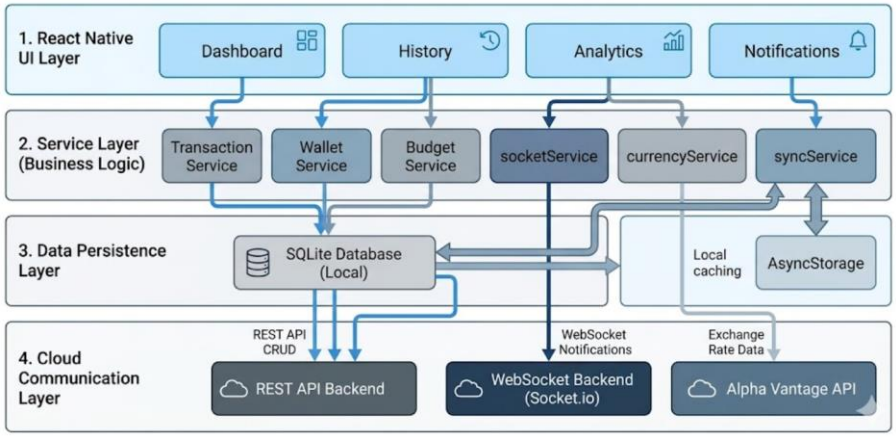
Users can:

- Set Preferred Currency: Choose between USD or MYR
- View Transactions: See all amounts in preferred currency
- Calculate Totals: Dashboard totals automatically use selected currency
- Maintain Accuracy: Conversion happens at calculation time using latest rates

Feature 3: Offline Fallback

- Last Known Rate: If API is unavailable, application uses last cached rate
- Graceful Degradation: Application continues functioning with cached data
- User Awareness: UI indicates when using cached (offline) rates
- Automatic Recovery: Retries fetching fresh rates when connectivity returns

C.2 Implementation of Cloud Connectivity



C.2.1 REST API Implementation

Configuration

The REST API base URL is configured in *appConfig.ts*:

- Supports both Android emulator and iOS simulator configurations
- Defaults to localhost for development
- Configurable via environment variables for different deployments
- Uses Flask backend running on port 5001

File: config/appConfig.ts

```
export const config = {
  appMode,
  isLocalMode: appMode === 'local',
  demoUserId: process.env.DEFAULT_USER_ID ?? 'demo-user',
  currencyCode: process.env.DEFAULT_CURRENCY_CODE ?? 'USD',
  sqliteDbName: process.env.SQLITE_DB_NAME ?? 'finance_tracker.sqlite',
  sqliteSeedDemoData: (process.env.SQLITE_SEED_DEMO ?? 'true') === 'true',
  apiBaseUrl:
    process.env.API_BASE_URL ??
    Platform.select({
      android: 'http://192.168.100.7:5001',
      ios: 'http://localhost:5001',
      default: 'http://localhost:5001',
    }),
};
```

URL for emulator and physical android device (Different IP address)

```
apiBaseUrl:
  process.env.API_BASE_URL ??
  Platform.select({
    android: 'http://10.0.2.2:5001',
    ios: 'http://localhost:5001',
    default: 'http://localhost:5001',
  }),
```

```
apiBaseUrl:
  process.env.API_BASE_URL ??
  Platform.select({
    android: 'http://192.168.100.7:5001',
    ios: 'http://192.168.0.6:5001',
    default: 'http://localhost:5001',
  }),
```

Authentication Header Management

Every REST API call includes Bearer token authentication:

- Retrieves JWT token from session (stored after login)
- Throws error if user not authenticated
- Used in all API calls to verify user identity
- Server validates token and returns 401 if invalid

File: services/transactionApi.ts

```
function getAuthHeaders(): Record<string, string> {
  const session = getAuthSession();
  if (!session?.token) {
    throw new Error('Please log in first');
  }

  return {
    Authorization: `Bearer ${session.token}`,
  };
}
```

Response Parsing with Error Handling

Error Handling Strategy:

- Checks HTTP status code
- Extracts error message from JSON payload
- Attaches status code for specific handling (e.g., 401 auth errors)
- Handles non-JSON responses gracefully

File: services/transactionApi.ts

```
async function parseResponse<T>(response: Response): Promise<T> {
  const contentType = response.headers.get('content-type') ?? '';
  const isJson = contentType.toLowerCase().includes('application/json');
  const payload = isJson ? await response.json() : null;

  if (!response.ok) {
    const message = payload?.message ?? `Request failed (${response.status})`;
    const err: any = new Error(message);
    // attach status so callers can react to auth errors (401)
    err.status = response.status;
    err.payload = payload;
    throw err;
  }

  return payload as T;
}
```

Network Error Detection:

- Distinguishes between network failures and application errors to implement appropriate retry logic.

File: services/transactionApi.ts

```
function isNetworkError(error: unknown): boolean {
  const message = String((error as any)?.message ?? '').toLowerCase();
  return message.includes('network request failed') || message.includes('failed to fetch');
}
```

Transaction Synchronization: The Complete Flow

Phase 1: Local Creation

When a user creates a transaction while offline:

1. Get Session: Retrieves authenticated user's ID
2. Insert Locally: Stores transaction in SQLite with temporary ID
3. Queue Change: Records operation in pending changes table
4. Update UI: Updates pending count badge
5. Trigger Sync: Initiates sync process (if online)
6. Return Immediately: User sees transaction instantly (optimistic update)

Phase 2: Continuous Sync Monitoring

The application runs a background sync every 15 seconds:

Sync Strategy:

- Sequential Processing: Handles changes one at a time in order
- FIFO Order: Processes oldest changes first (maintains causality)
- Atomic Operations: Each change completes fully before moving to next
- Error Tolerance: Stops on network errors (retries later), also stops on server errors

Phase 3: CREATE Operation Detail

Process:

1. POST to */api/transactions* with transaction details'
2. Server generates permanent ID and stores in database

3. Client receives ID and updates all local references (important for wallets)

4. Removes operation from pending queue

5. Continues to next pending operation

Phase 4: UPDATE and DELETE Operations

Key Differences:
- UPDATE: Uses PUT method, only sends changed fields
- DELETE: Uses DELETE method, no body required
- Both receive confirmation and remove from pending queue

Phase 5: Sync Timer Management

Timing:
- Starts immediately when called
- Runs every 15 seconds (15000 ms)
- Can be stopped to prevent unnecessary network activity
- Typically started on app launch and stopped on logout

Data Fetching and Merging

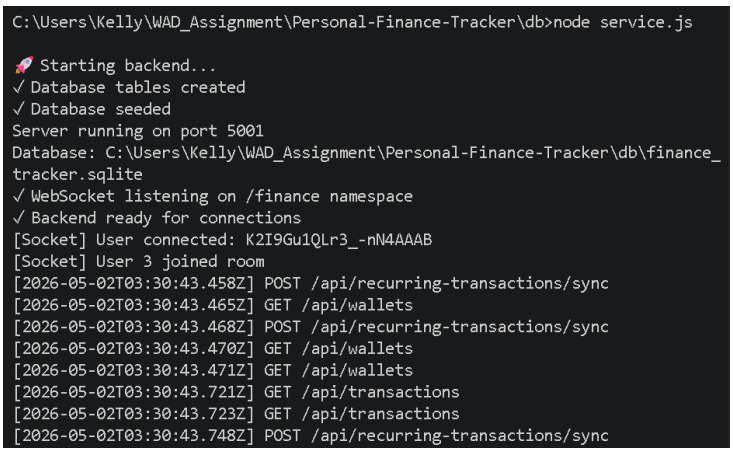
Fetching Transactions with Merge Strategy
The Merge Algorithm

C.2.2 WebSocket Implementation

Content

C.2.3 Currency Exchange Rate Integration

Content


C:\Users\Kelly\WAD_Assignment\Personal-Finance-Tracker\db>node service.js
🚀 Starting backend...
✓ Database tables created
✓ Database seeded
Server running on port 5001
Database: C:\Users\Kelly\WAD_Assignment\Personal-Finance-Tracker\db\finance_
tracker.sqlite
✓ WebSocket listening on /finance namespace
✓ Backend ready for connections
[Socket] User connected: K2I9Gu1QLr3_-nN4AAAB
[Socket] User 3 joined room
[2026-05-02T03:30:43.458Z] POST /api/recurring-transactions/sync
[2026-05-02T03:30:43.465Z] GET /api/wallets
[2026-05-02T03:30:43.468Z] POST /api/recurring-transactions/sync
[2026-05-02T03:30:43.470Z] GET /api/wallets
[2026-05-02T03:30:43.471Z] GET /api/wallets
[2026-05-02T03:30:43.721Z] GET /api/transactions
[2026-05-02T03:30:43.723Z] GET /api/transactions
[2026-05-02T03:30:43.748Z] POST /api/recurring-transactions/sync

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